

KAVISHKA DULSHAN | *B.Sc. (Hons) in Software Engineering (Undergraduate)*

GitHub - <https://github.com/KavishkaDulshan/>

LinkedIn - <https://www.linkedin.com/in/kavishka-dulshan/>

Website - <https://kavishkadulshan.dev/>

Email - kavishkadulshan2@gmail.com

Mobile: +94 712 815 314 / **WhatsApp:** +94 78 582 6121

Address: Kalutara, Sri Lanka

OBJECTIVE

A motivated Software Engineering undergraduate with a strong passion for continuous learning through hands on building. Experienced in developing full stack software, IoT systems, and AI driven applications. Driven by a natural curiosity for how technologies operate under the hood, combining solid technical execution with effective collaboration and teamwork.

EDUCATION

- **B.Sc. (Hons) in Software Engineering (2024 – Present)**
 - National School of Business Management – Faculty of Computing (NSBM Green University)
 - **CGPA:** 3.81 (4 semesters)
 - **Semester GPA:** (Semester 1 – 3.86, Semester 2 – 4.0, Semester 3 – 3.66, Semester 4 – 3.74)
- **GCE Advanced Level Examination (2021) – Commerce Stream**
 - Kalutara Vidyalaya National School, Kalutara
 - ICT, Accounting, Economics
- **Diploma in English – Britishway English Academy (2019)**

PROJECTS

1. **Online Auction System Backend** -Architected and developed a scalable .NET 8 Web API utilizing an N Tier architecture and the Repository Pattern to decouple data access from business logic. Implemented a real time bidding engine via SignalR WebSockets to manage concurrent bidding rooms and broadcast instant updates. Engineered an automated background service for lifecycle transitions, multipart file uploads with FluentValidation, and robust security through JWT authentication. [Read more](#)
2. **BlinkChat** – Developed a cross platform messaging application using Flutter, Node.js, and Socket.io for sub-second text and media delivery, backed by a containerized MongoDB instance. Implemented Multi-Factor Authentication (MFA), Firebase Push Notifications, and granular security features including Helmet.js, rate limiting, and NoSQL injection defenses. Automated deployment to Azure via GitHub Actions and am currently migrating the codebase from a monolithic architecture to a scalable microservices architecture to support heavy traffic spikes. [Read more](#)
3. **Go4 (Multimodal AI Retail Navigator)** – An AI powered retail mobile app built using Flutter, Node.js, Express, and MongoDB. Integrates Gemini and OpenAI APIs for multimodal (text,voice and image) search and personalized recommendations, deploying a containerized backend orchestrated via automated GitHub Actions CI/CD pipelines. [Read more.](#)
4. **Smart Companion Robot** – An interactive Edge AI desktop companion programmed with C++, ESP32 S3, and Edge Impulse. Features MPU6050 motion tracking and a custom machine learning model achieving around 80% on device keyword recognition accuracy. Additionally, it connects to a custom-built Android application over the internet, allowing users to control and customize the robot remotely. [Read more.](#)
5. **Restaurant POS System** – A local, offline first Point of Sale application built using Flutter, Node.js, PostgreSQL, and Docker. Features "Smart KOT" logic to intelligently route orders and uses Docker to simulate hardware execution for 100+ daily orders. [Read more.](#)
6. **Restaurant Management System (Inventory Module)**- Developed a mission-critical inventory subsystem using Jakarta EE 10, Java 22, and Hibernate ORM to automate stock tracking and reduce food waste. Structured a robust MVC framework with Data Access Objects (DAOs) to execute transaction logic and queries against a MySQL database. Created an intuitive web dashboard featuring custom CSS components, jBCrypt security, and real time usage forms to optimize daily operation efficiency. [Read more](#)

7. [Brand Portfolio](#) – A high end, visual first digital portfolio built with React 19, Vite, and Tailwind CSS. Engineered modular component architecture and a custom UI design system to effectively showcase curated product lines and interior design services. [Read more.](#) | [Live demo](#)
8. [Relper \(Read Helper\)](#) – Created a cross-platform, open-source PDF reading assistant using React 19, Vite, and pdfjs-dist to streamline document parsing for non-native English readers. Engineered instant dictionary lookup utilities and built a resilient fallback pipeline with Pollinations AI to provide zero-cost text explanations. Delivered a lightweight desktop application using a Python Flask backend and pywebview wrapped into a standalone executable via PyInstaller. [Read more](#) | [Live demo](#)
9. [Restricted Video Downloader](#) - Built a custom Chrome DevTools Extension utilizing Manifest V3 to bypass progressive video download restrictions through network request interception. Engineered byte perfect file reassembly by parsing HTTP Content Range headers and utilizing JavaScript Uint8Array buffers to prevent binary data corruption during Base64 decoding. Constructed a live DevTools UI panel to track download progress and automatically trigger native browser Blob exports. [Read more.](#)

TECHNICAL SKILLS

- **Languages** – JavaScript, Python, Dart, Java, C++, C#
- **Core Concepts** - Data Structures & Algorithms, SDLC , ADBMS
- **Frontend & Web Development** - React, Flutter, Tailwind CSS
- **Backend & APIs** - Node.js (Express), .NET 8 (C#), Jakarta EE 10, Python (Flask), RESTful APIs
- **Databases & Infrastructure** - MySQL, MongoDB, PostgreSQL, Azure, Docker, GitHub Actions
- **Hardware & Edge AI** - ESP32 S3, Raspberry Pi, Edge Impulse (TinyML), UART
- **APIs & Services** – Firebase, Gemini API, Groq, Serper.dev
- **Version Controlling** - Git, GitHub

SOFT SKILLS

- **Leadership & Communication** – Coordinated and led several university project teams of 6 -10 members.
- **Mentoring & Tutoring** – Tutored Ordinary Level (O/L) and Advanced Level (A/L) students in ICT, simplifying complex technical concepts.
- **Problem Solving & Initiative** – Autonomously managed and successfully delivered real world freelance and academic projects.
- **Adaptability** – Versatile and effective at bridging the gap between hardware development and software architecture.

CERTIFICATIONS

- **Web Design for Beginners** - University of Moratuwa (July 2025)
- **Python for Beginners** - University of Moratuwa (September 2023)
- **EF SET English Certificate** - 67/100 (C1 Advanced) (June 2024)
- **From Relational Model (SQL) to MongoDB's Document Model** (March 2026)

EXTRA CURRICULAR ACTIVITIES

- **Blog Writer** - Research and publish articles focusing on software development, modern web tools, IOT and technical architectures, simplifying complex programming concepts for readers.
- **Tutor for O/L and A/L ICT students** (June 2024-June 2026)- Mentoring and tutoring Ordinary Level and Advanced Level students in ICT, designing simplified lesson structures for complex algorithmic and logical concepts.
- **Member** – FOSS Community, NSBM Green University

REFERENCE

Mr. Dinesh De Saram
Business Development Manager
Hemas Pharmaceuticals
Colombo 03.
077 418 2809 / 075 4766666

Ms. Pavithra Subashini
Senior lecturer - Department of Software
Engineering - Faculty of Computing - NSBM
Green University
077 618 1390
pavithras@nsbm.ac.lk